

Sample Complexity of Composite Quantum Hypothesis Testing

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Outline

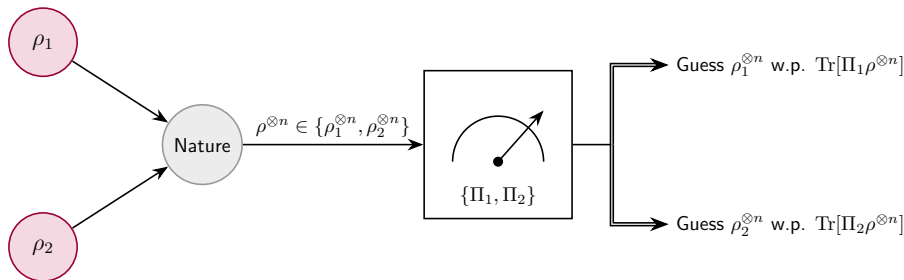
- 1 Simple QHT
- 2 Our Problem Setting: Composite QHT
- 3 The Sample Complexity of Composite QHT
- 4 Extension to Differentially Private Composite QHT
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What is Simple QHT?

- Given n copies of an unknown quantum state $\rho^{\otimes n} \in \{\rho_1^{\otimes n}, \rho_2^{\otimes n}\}$, how can we decide which state was prepared?
- We perform a binary POVM measurement $\{\Pi_1, \Pi_2\}$, where outcome i corresponds to deciding in favor of $\rho_i^{\otimes n}$.



- Type I Error:** Probability of guessing $\rho_2^{\otimes n}$ when $\rho_1^{\otimes n}$ is true is $\text{Tr}[\Pi_2 \rho_1^{\otimes n}]$.
- Type II Error:** Probability of guessing $\rho_1^{\otimes n}$ when $\rho_2^{\otimes n}$ is true is $\text{Tr}[\Pi_1 \rho_2^{\otimes n}]$.

Sample Complexity of Simple QHT

- The distinguisher wishes to design the optimal POVM which minimizes the expected probability of error:

$$P_{e,\min}(p, \rho_1^{\otimes n}, \rho_2^{\otimes n}) := \inf_{\Pi_1 + \Pi_2 = I, \Pi_i \succeq 0} p \operatorname{Tr} [\Pi_2 \rho_1^{\otimes n}] + (1 - p) \operatorname{Tr} [\Pi_1 \rho_2^{\otimes n}],$$

where $p \in (0, 1)$ is the prior probability for preparing state ρ_1 .

Definition of Sample Complexity

For $\delta \in (0, 1)$, the **sample complexity** is,

$$n^*(p, \rho_1, \rho_2, \delta) := \inf\{n \in \mathbb{N} : P_{e,\min}(p, \rho_1^{\otimes n}, \rho_2^{\otimes n}) \leq \delta\}. \quad (1)$$

That is the minimum number of copies of the unknown state required such that $P_{e,\min}(p, \rho_1^{\otimes n}, \rho_2^{\otimes n}) \leq \delta$ for some error threshold $\delta \in (0, 1)$.

Sample Complexity of Simple QHT (Cheng *et al.*, 2025)

The following bounds hold for simple QHT:

$$\frac{\ln\left(\frac{p(1-p)}{\delta}\right)}{-2 \ln F(\rho_1, \rho_2)} \leq n^*(p, \rho_1, \rho_2, \delta) \leq \left\lceil \inf_{s \in [0,1]} \frac{\ln\left(\frac{p^s(1-p)^{1-s}}{\delta}\right)}{-\ln \text{Tr}(\rho_1^s \rho_2^{1-s})} \right\rceil.$$

Specifically, for fixed prior p , the sample complexity is characterized as

$$n^*(p, \rho_1, \rho_2, \delta) = \Theta\left(\frac{\ln \frac{1}{\delta}}{-\ln F(\rho_1, \rho_2)}\right).$$

- The (square-root) Uhlmann fidelity is

$$F(\rho_1, \rho_2) := \|\sqrt{\rho_1}\sqrt{\rho_2}\|_1,$$

where $F(\rho_1, \rho_2) \in [0, 1]$ is a measure of ‘similarity’ between the states.

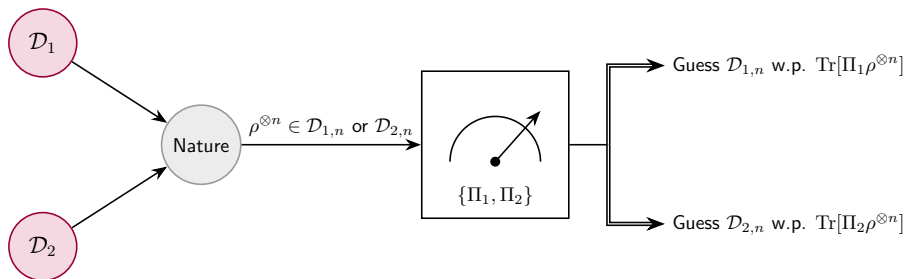
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Our Problem Setting: Composite QHT

- In composite QHT, the n -copy unknown quantum state $\rho^{\otimes n}$ belongs to one of two **compact** uncertainty sets, $\mathcal{D}_{1,n}$ or $\mathcal{D}_{2,n}$, where $\rho \in \mathcal{S}(\mathbb{C}^d)$.
- We assume the n -copy unknown quantum state is i.i.d.

$$\mathcal{D}_{1,n} := \{\rho_1^{\otimes n} : \rho_1 \in \mathcal{D}_1\} \text{ and } \mathcal{D}_{2,n} := \{\rho_2^{\otimes n} : \rho_2 \in \mathcal{D}_2\}$$



Our Main Goal

- The optimal min-max expected error probability is

$$P_{e,\min}(p, \mathcal{D}_{1,n}, \mathcal{D}_{2,n}) := \inf_{\Pi_1 + \Pi_2 = I, \Pi_i \succeq 0} \sup_{\substack{\rho_1^{\otimes n} \in \mathcal{D}_{1,n} \\ \rho_2^{\otimes n} \in \mathcal{D}_{2,n}}} p \operatorname{Tr} [\Pi_2 \rho_1^{\otimes n}] + (1-p) \operatorname{Tr} [\Pi_1 \rho_2^{\otimes n}].$$

Definition of Composite Sample Complexity

For $\delta \in (0, 1)$ the **composite sample complexity** is,

$$n^*(p, \mathcal{D}_1, \mathcal{D}_2, \delta) := \inf \{n \in \mathbb{N} : P_{e,\min}(p, \mathcal{D}_{1,n}, \mathcal{D}_{2,n}) \leq \delta\}. \quad (2)$$

That is the minimum number of copies of the unknown state required such that $P_{e,\min}(p, \mathcal{D}_{1,n}, \mathcal{D}_{2,n}) \leq \delta$ for some error threshold $\delta \in (0, 1)$.

Our goal is to derive lower and upper bounds on (2).

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Upper Bounds: Preliminaries

Preliminary Result

The following bound holds:

$$P_{e,\min}(p, \mathcal{D}_{1,n}, \mathcal{D}_{2,n}) \leq \sqrt{p(1-p)} \sup_{\sigma_{i,n} \in \mathcal{C}_{i,n}, i=1,2} F(\sigma_{1,n}, \sigma_{2,n}),$$

where $\mathcal{C}_{i,n}$ is the **convex hull** of $\mathcal{D}_{i,n}$, i.e. $\mathcal{C}_{i,n} := \text{conv}(\mathcal{D}_{i,n})$.

Why do we have convex hulls?

$\rho \rightarrow \rho^{\otimes n}$ is non-linear, so $\mathcal{D}_{i,n}$ is not convex even if $\mathcal{D}_i$ is convex!

Proof (Sketch)

- The minimax error is equal to the optimization over the convex hulls of the uncertainty sets.
- Use Sion's minimax theorem to swap the sup and inf and applying the Holevo–Helstrom theorem gives,

$$P_{e,\min}(p, \mathcal{D}_{1,n}, \mathcal{D}_{2,n}) = \sup_{\sigma_{i,n} \in \mathcal{C}_{i,n}, i=1,2} \left(\frac{1}{2} - \frac{1}{2} \|p\sigma_{1,n} - (1-p)\sigma_{2,n}\|_1 \right).$$

- Use the Quantum Chernoff Bound (Audenaert *et al.*, 2007) with $s = 1/2$ and use $\text{Tr}[\sqrt{\sigma_{1,n}}\sqrt{\sigma_{2,n}}] \leq \|\sqrt{\sigma_{1,n}}\sqrt{\sigma_{2,n}}\|_1 = F(\sigma_{1,n}, \sigma_{2,n})$. ■

Upper Bounds: General

General Upper Bound

Let $\mathcal{D}_1, \mathcal{D}_2$ be compact sets of quantum states satisfying $F_{\max} \leq c$ for some $c \in (0, 1)$, where

$$F_{\max} := \sup_{\rho_i \in \mathcal{D}_i, i=1,2} F(\rho_1, \rho_2)$$

is the maximum fidelity between the uncertainty sets. Then for $p, \delta \in (0, 1)$

$$n^*(p, \mathcal{D}_1, \mathcal{D}_2, \delta) \leq \left\lceil \frac{2 \ln \left(\frac{\sqrt{p(1-p)} K_{c,d}}{\delta} \right)}{-\ln F_{\max}} \right\rceil, \quad (3)$$

where $K_{c,d} := \binom{N+d^2-1}{N} c^{N/2}$ and $N := \max \left\{ 1, \left\lceil \frac{d^2 \sqrt{c}-1}{1-\sqrt{c}} \right\rceil \right\}$.

Proof (Sketch)

- Recall that

$$P_{e,\min}(p, \mathcal{D}_{1,n}, \mathcal{D}_{2,n}) \leq \sqrt{p(1-p)} \sup_{\sigma_{i,n} \in \mathcal{C}_{i,n}, i=1,2} F(\sigma_{1,n}, \sigma_{2,n}).$$

Upper Bounds: General

Proof (Sketch)

- By Carathéodory's theorem, write $\sigma_{i,n}$ as $\dim(\mathcal{C}_{i,n}) + 1$ states from $\mathcal{D}_{i,n}$.
- Then using subadditivity of fidelity, Cauchy-Schwarz and $\dim(\mathcal{C}_{i,n}) + 1 \leq \binom{n+d^2-1}{n}$, we get

$$P_{e,\min}(p, \mathcal{D}_{1,n}, \mathcal{D}_{2,n}) \leq \sqrt{p(1-p)} \overbrace{\binom{n+d^2-1}{n}}^{\text{not monotonically decreasing in } n} F_{\max}^n.$$

- Lastly, since $F_{\max} \leq c$, we have

$$\begin{aligned} P_{e,\min}(p, \mathcal{D}_{1,n}, \mathcal{D}_{2,n}) &\leq \sqrt{p(1-p)} \binom{n+d^2-1}{n} c^{n/2} F_{\max}^{n/2} \\ &\leq \sqrt{p(1-p)} K_{c,d} F_{\max}^{n/2}. \end{aligned}$$

- The choice of n in (3) satisfies $P_{e,\min}(p, \mathcal{D}_{1,n}, \mathcal{D}_{2,n}) \leq \delta$, completing the proof. ■

The Lower Bound

Lower Bound

Let $\mathcal{D}_1, \mathcal{D}_2$ be arbitrary compact sets of quantum states. Then, for $p, \delta \in (0, 1)$

$$\frac{\ln \frac{p(1-p)}{\delta}}{-2 \ln F_{\max}} \leq n^*(p, \mathcal{D}_1, \mathcal{D}_2, \delta). \quad (4)$$

Additionally, suppose that $F_{\max} \leq c$ for some $c \in (0, 1)$. Then for fixed p, c and d , the sample complexity scales as

$$n^*(\delta) = \Theta\left(\frac{\ln \frac{1}{\delta}}{-\ln F_{\max}}\right).$$

Proof

- For any pair $(\rho_1, \rho_2) \in \mathcal{D}_1 \times \mathcal{D}_2$, we have $n^*(p, \mathcal{D}_1, \mathcal{D}_2, \delta) \geq n^*(p, \rho_1, \rho_2, \delta)$.
- Thus, $n^*(p, \mathcal{D}_1, \mathcal{D}_2, \delta) \geq \sup_{\rho_i \in \mathcal{D}_i} n^*(p, \rho_1, \rho_2, \delta)$.
- The result follows from taking the supremum of the simple lower bound. ■

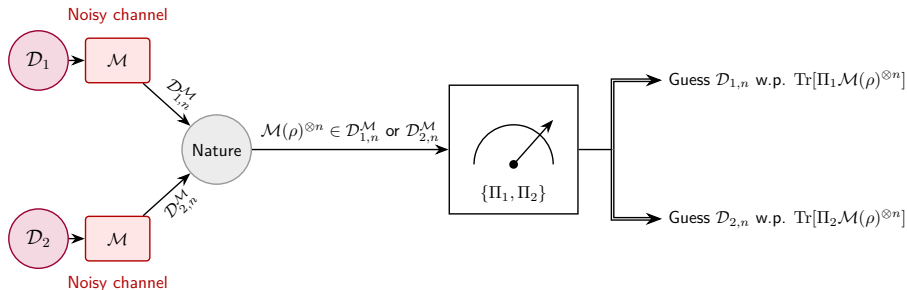
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Differentially Private Composite QHT

- The locally private uncertainty sets are:

$$\mathcal{D}_i^{\mathcal{M}} := \{\mathcal{M}(\rho_i) : \rho_i \in \mathcal{D}_i\} \text{ and } \mathcal{D}_{i,n}^{\mathcal{M}} := \{\mathcal{M}(\rho_i)^{\otimes n} : \rho_i \in \mathcal{D}_i\}$$



ε -Locally Differentially Private Quantum Channel

A CPTP map $\mathcal{M}$ is ε -LDPQ for $\varepsilon \geq 0$ if

$$\sup_{\rho_1, \rho_2} E_{e^\varepsilon}(\mathcal{M}(\rho_1) \parallel \mathcal{M}(\rho_2)) = 0, \quad (5)$$

where $E_\gamma(\rho_1 \parallel \rho_2) := \text{Tr}[(\rho_1 - \gamma \rho_2)_+]$ is the quantum hockey-stick divergence.

LDPQ Sample Complexity

For $\varepsilon \geq 0$ and $\delta \in (0, 1)$ the **LDPQ sample complexity** is,

$$n_{\varepsilon}^*(p, \mathcal{D}_1, \mathcal{D}_2, \delta) := \inf_{\mathcal{M} \in \text{LDP}_{\varepsilon}} n^*(p, \mathcal{D}_1^{\mathcal{M}}, \mathcal{D}_2^{\mathcal{M}}, \delta), \quad (6)$$

where LDP_{ε} is the set of all ε -LDPQ channels.

- The minimal quantum hockey-stick divergence (for $\gamma = 1$) between the uncertainty sets is

$$E_{1,\min} := \inf_{\rho_i \in \mathcal{D}_i, i=1,2} E_1(\rho_1 \| \rho_2) = \inf_{\rho_i \in \mathcal{D}_i, i=1,2} \frac{1}{2} \|\rho_1 - \rho_2\|_1$$

Differentially Private Sample Complexity Bounds

LDPQ Sample Complexity Bounds

Let $\mathcal{D}_1, \mathcal{D}_2$ be compact sets of quantum states, possibly of infinite cardinality, satisfying $F_{\max} \leq c$ for some $c \in (0, 1)$. For $\varepsilon \geq 0$ and $p, \delta \in (0, 1)$, we have

$$\frac{e^\varepsilon}{(e^\varepsilon - 1)^2} \frac{(1 - \frac{\delta}{p})^2}{(E_{1,\min})^2} \leq n_\varepsilon^*(p, \mathcal{D}_1, \mathcal{D}_2, \delta) \leq \left\lceil \left(\frac{e^\varepsilon + 1}{e^\varepsilon - 1} \right)^2 \frac{4 \ln \left(\frac{K_{c,d} \sqrt{p(1-p)}}{\delta} \right)}{(E_{1,\min})^2} \right\rceil$$

where $K_{c,d} := \binom{N+d^2-1}{N} c^{N/2}$ and $N := \max \left\{ 1, \left\lceil \frac{d^2 \sqrt{c}-1}{1-\sqrt{c}} \right\rceil \right\}$. Additionally, for fixed p, δ, c and d , the sample complexity scales as

$$\Omega \left(\frac{e^\varepsilon}{(e^\varepsilon - 1)^2} \frac{1}{(E_{1,\min})^2} \right) \leq n_\varepsilon^*(p, \mathcal{D}_1, \mathcal{D}_2, \delta) \leq \mathcal{O} \left(\left(\frac{e^\varepsilon + 1}{e^\varepsilon - 1} \right)^2 \frac{1}{(E_{1,\min})^2} \right).$$

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Summary of Results

Setting	Lower Bound	Upper Bound
Simple [1]	$\frac{\ln \frac{p(1-p)}{\delta}}{-2 \ln F(\rho_1, \rho_2)}$	$\left[\inf_{s \in [0,1]} \frac{\ln \frac{p^s(1-p)^{1-s}}{\delta}}{-\ln \text{Tr}(\rho_1^s \rho_2^{1-s})} \right]$
Composite	$\frac{\ln \frac{p(1-p)}{\delta}}{-2 \ln F_{\max}}$	$\left[\frac{2 \ln \frac{\sqrt{p(1-p)} K_{c,d}}{\delta}}{-\ln F_{\max}} \right]$ $(F_{\max} \leq c)$
Private Simple [2] $(p = \frac{1}{2}, \delta = 0.1)$	$\frac{16}{25} \frac{e^\varepsilon}{(e^\varepsilon - 1)^2 E_1(\rho_1 \ \rho_2)^2}$	$\left(\frac{e^\varepsilon + 1}{e^\varepsilon - 1} \right)^2 \frac{2 \ln 5}{E_1(\rho_1 \ \rho_2)^2}$
Private Composite $(p = \frac{1}{2}, \delta = 0.1)$	$\frac{16}{25} \frac{e^\varepsilon}{(e^\varepsilon - 1)^2 (E_{1,\min})^2}$	$\left[\left(\frac{e^\varepsilon + 1}{e^\varepsilon - 1} \right)^2 \frac{4 \ln(5 K_{c,d})}{(E_{1,\min})^2} \right]$ $(F_{\max} \leq c)$

[1] Cheng et al., *An Invitation to the Sample Complexity of Quantum Hypothesis Testing*.

[2] Cheng et al., *Sample Complexity of Locally Differentially Private Quantum Hypothesis Testing*.

Conclusion & Future work

- Characterized the sample complexity of symmetric composite QHT,

$$n^*(\delta) = \Theta\left(\frac{\ln(\frac{1}{\delta})}{-\ln F_{\max}}\right),$$

- Extended the framework to the LDPQ setting.

- **Future work:**

- Asymmetric composite QHT.
- M -ary composite QHT.
- Non-i.i.d. QHT:

$$\mathcal{D}_i^{\otimes n} = \left\{ \bigotimes_{j=1}^n \rho_{i,j} : \rho_i \in \mathcal{D}_i \right\}.$$